

AMAN AHMED

+1-510-565-9897 | amanahmed212@gmail.com | [in amanahmed212](https://www.linkedin.com/in/amanahmed212) | [aman1678](https://github.com/aman1678) | San Francisco Bay Area, CA

U.S. Citizen

OBJECTIVE

Physics graduate with hands-on software development experience across full-stack applications, REST APIs, and numerical simulation, seeking an entry-level Software Engineer role on a cloud storage team. Comfortable writing clean, tested, maintainable code, working with Git and collaborative code review, and picking up new languages and systems quickly. Eager to build a foundation in storage protocols and distributed computing while contributing well-scoped features under the guidance of senior engineers.

EDUCATION

- **University of California, San Diego** Graduated June 2025
B.S. in Physics with Honors La Jolla, CA
 - Certifications: CompTIA A+ (Core 1 Complete, Core 2 In Progress) – Linux fundamentals, computer architecture, and hardware/software troubleshooting
 - Coursework: Computational Physics, Circuits and Electronics, Quantum Mechanics, Electrodynamics, Advanced Optics

EXPERIENCE

- **Full Stack Developer (Freelance)** February 2025 – June 2025
Qanvas (UC San Diego Startup) La Jolla, CA
 - Implemented and debugged well-scoped features across a full-stack application built with **React**, **HTML**, **CSS**, and **JavaScript**, using **Git** for source control and coordinating changes with teammates through code review.
 - Built and maintained 10+ **REST API** endpoints spanning authentication, data management, and service-to-service communication, testing and troubleshooting functionality end-to-end.
 - Maintained a **PostgreSQL** database and wrote clear technical documentation of schema changes and design decisions, reinforcing habits around clean, maintainable, well-documented code.
- **Math Tutor** March 2023 – June 2023
Kumon Mountain House, CA
 - Diagnosed individual gaps in quantitative reasoning and coordinated targeted solutions with center staff, building root-cause analysis skills applicable to debugging and troubleshooting.
 - Translated complex mathematical and physical concepts into clear, accessible explanations, demonstrating strong written and verbal communication across audiences.
- **Line Cook** August 2022 – December 2022
THub Cafe Mountain House, CA
 - Coordinated closely with a team under time pressure, communicating proactively to keep multiple concurrent tasks moving smoothly and accurately.

PROJECTS

- **Gravitational Simulation Engine**
Python, Numerical Methods, Systems Modeling
 - Designed and implemented a numerical simulation engine modeling gravitational N-body dynamics, choosing integration schemes to balance accuracy and computational performance.
 - Wrote validation tests checking simulation output against known analytical solutions and physical conservation laws, developing a rigorous approach to testing and verifying system behavior.
- **Drama Review Platform**
Python, FastAPI, SQLite, React, REST API, SQLAlchemy, Git
 - Built and maintained a full-stack application end-to-end, implementing and testing features across the **React** frontend, **FastAPI** backend, and **SQLite** database using **Git** for version control.
 - Documented API endpoints and database schema decisions to create a clear knowledge base, supporting maintainability and onboarding.

SKILLS

- **Languages:** Python, JavaScript, SQL; comfortable picking up Java/Go
- **Systems & Tools:** Git, Linux Fundamentals, REST APIs, Bash, Basic CI/CD Concepts
- **Software Practices:** Testing & Validation, Code Review, Technical Documentation, Clean & Maintainable Code
- **Foundations:** Data Structures, Numerical Methods, Networking Fundamentals, Distributed Systems (coursework/self-study)
- **Soft Skills:** Cross-Functional Communication, Collaborative Mindset, Problem-Solving, Attention to Detail

HONORS

- **Tau Sigma National Honors Society** February 2024
 - Recognized for academic excellence and leadership among transfer students.